

Capstone Project Preliminary Analysis Template

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Project Title: CAPSTONE: RETAIL ANALYTICS-PREDICTING CUSTOMER PURCHASE BEHAVIOR

Project Objective: *What is the goal? What problem are you trying to solve? Who will be your audience?*

Business Introduction: RetailVC's Retail Analytics Solutions, a leading provider of data-driven solutions for retail businesses. RetailVC specializes in using advanced analytics techniques to help retailers optimize their operations, enhance customer experiences, and drive business growth. Its mission is to empower retail organizations with actionable insights that enable them to make informed decisions, improve performance, and remain competitive in today's dynamic marketplace.

Problem Statement: The goal of this capstone project is to develop a comprehensive understanding of customer behavior and preferences based on retail transaction data. The aim is to analyze customers' demographic information, purchasing patterns, and interactions with marketing campaigns to identify key insights. These insights will inform strategic decision-making and drive business growth. Specifically, the focus will be on predicting customers' responses to marketing campaigns and optimizing promotional strategies to maximize their effectiveness and return on investment (ROI).

Data Collection Source: *Describe how the data was collected. Include website links.*

The datasets were collected and provided by FULLSTACK ACADEMY

Google drive link:

https://drive.google.com/drive/folders/12_1UqBRGjXit3RAc6pTS5rqMHD3IB1QE

Methodology: *Describe your preprocessing techniques. How was the data organized? What was the workflow? What tools were used to organize and better understand the data? How was the data cleaned? Were there any typos? How was this corrected? Any data type conversions, e.g. string to date values? Any other formatting, e.g. renaming columns or values? Were any missing values replaced? If so, describe why? How will replacing the values impact the analysis? Are*

there any exclusions (columns, rows, or values)? Include any other methods in processing and organizing the data.

Workflow:

- Read, digest, and understand what is being asked of me.
- Read through the business goal/objective and brainstorm final product visualization in order to build a relationship with the story telling of the datasets
- Download datasets. In this case, it was provided in the Google Drive link above.
- Import in Excel
- Within Excel, I make sure there is a Data Dictionary table, a Findings table, and an overall Summary sheet – this will allow me to stay organized with my workflow
- Notetaking
- With the loaded datasets, I am first making a working copy in order to protect the original data.
- Prep the data- check datatypes. Create a Metadata sheet that will have the headers, the data types, and the description of the column.
- Formatting the table prior to further analysis.
- Data cleaning: Cell height and widths, dates (utilizing SQL to format the dates), any duplicates, missing values, blanks, spelling errors, sorting the data
- Save the worksheet, export and import into DBeaver for further SQL analysis.

Tools:

- Excel- exploring the data and very basic data cleaning/prepping. Understanding the goal of the project. The objectives, the impact, the importance.
- SQL -further data cleaning and formatting. Building database in order to dive more into data analysis.

Findings from data to stay curious about:

- Column: Birthyear, ranges from years 1892-1996
- Column: Enrollment dates, years: 2012, 2013, 2014
- Column: Education- what's the significance of education to customer purchase behavior
- Column: Marital Status, what's the significance of this to customer purchase behavior

- Column: Kidhome and teenhome, what's the significance of this to customer purchase behavior

Findings:

- 1. Data Size:** *How many rows and columns?*
- 2. Missing Values:** *How many? Will these need to be replaced or deleted?*
- 3. Duplicates:** *Describe how this was detected*
- 4. Dates:** *What is the date range? Will this need to be modified for the project?*
- 5. Additional Findings:** *Identify any other findings or errors in the data. Any outliers? Any inconsistencies relative to the project?*

DATA DICTIONARY

Attributes	Description	Updated column name
People		
ID	Customer's unique identifier	ID
Year_Birth	Customer's birth year	BirthYear
Education	Customer's education level	Education
Marital_Status	Customer's marital status	MaritalStatus
Income	Customer's yearly household income	Income
Kidhome	Number of children in customer's household	Kidhome
Teenhome	Number of teenagers in customer's household	Teenhome
Dt_Customer	Date of customer's enrollment with the company	Enrolldate
Recency	Number of days since customer's last purchase	Recency
Complain	1 if the customer complained in the last 2 years, 0 otherwise	Complain
Products		
MntWines	Amount spent on wine in last 2 years	AmntWines

MntFruits	Amount spent on fruits in last 2 years	AmntFruits
MntMeatProducts	Amount spent on meat in last 2 years	AmntMeatProducts
MntFishProducts	Amount spent on fish in last 2 years	AmntFishProducts
MntSweetProducts	Amount spent on sweets in last 2 years	AmntSweetProducts
MntGoldProds	Amount spent on gold in last 2 years	AmntGoldProds
Promotion		
NumDealsPurchases	Number of purchases made with a discount	NumDsetPurchases
AcceptedCmp1	1 if customer accepted the offer in the 1st campaign, 0 otherwise	AcceptedCmp1
AcceptedCmp2	1 if customer accepted the offer in the 2nd campaign, 0 otherwise	AcceptedCmp2
AcceptedCmp3	1 if customer accepted the offer in the 3rd campaign, 0 otherwise	AcceptedCmp3
AcceptedCmp4	1 if customer accepted the offer in the 4th campaign, 0 otherwise	AcceptedCmp4
AcceptedCmp5	1 if customer accepted the offer in the 5th campaign, 0 otherwise	AcceptedCmp5
Response	1 if customer accepted the offer in the last campaign, 0 otherwise	Response
Place		
NumWebPurchases	Number of purchases made through the company's website	NumWebPurchases
NumCatalogPurchases	Number of purchases made using a catalogue	NumCatalogPurchases
NumStorePurchases	Number of purchases made directly in stores	NumStorePurchases
NumWebVisitsMonth	Number of visits to company's website in the last month	NumWebVisitsMonth

TASKS

Rows	2241
Columns	27
Count:	60483

Blanks	24 (under income column)
Duplicates	0
Spelling Errors	0

Formatting:

sort dataset by ID smallest to largest
added filter tool on the columns
USD (\$) format for income
Formatted the Enrollment Date column to MM/DD/YYYY using SQL
Cell height and width adjusted
Rearranged columns for better understanding of the dataset
Personalized column names/headers from OG dataset

Attach any SQL or Python code in a separate file